

Deep Learning-Based Fruit Freshness Assessment with ResNet34 and Explainable Visualizations

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Abstract—Freshness of fruit is a very important factor in determining the food quality and food safety as well as the commercial value of fruit, but the conventional food inspection procedures are mostly manual, subjective, and lack uniformity. As the production demand heightens and quality control is required with efficiency in agricultural and retail industry. One of the benefits of automatic evaluation of the freshness of fruits is the reduction of labor, subjectivity, and imprecision of the manual analysis. But current research is commonly not well-validated, interpretable, and computationally efficient, which restricts its applicability in the real world. This research fills these gaps by comparing state-of-the-art transformers and transfer learning-based models using an improved methodology on a six-class large dataset. Among all architectures, ResNet34 achieved the best performance with 99.93% test accuracy, excellent cross-validation consistency, with a mean value of 0.9961, and the highest kappa value. Grad-CAM visualizations also revealed that predictions were based on freshness-related meaningful features. The high accuracy, interpretability, and computational efficiency indicate that the suggested ResNet34-based framework is perfectly adapted to the realistic, automated fruit freshness evaluation.

Index Terms—Fruit freshness classification, Transfer learning, ResNet34, Grad-CAM, PSNR, SSIM.

I. INTRODUCTION

Freshness of fruits is one of the primary factors defining the quality and safety of food and commercial value, therefore, it is a crucial factor in the agricultural and retail industry [1]. Historically, the evaluation of fruit freshness is based on manual inspection that is time-consuming, involves a lot of labor and subject to human error. Manual sorting tends to be subjective and lack consistency, especially in cases where the sorting is to be done on nuanced decay signs, which may not be noticeable to the eye. With rising demand for high-quality produce and effective quality control, there is a growing need to have automated and reliable fruit freshness evaluation processes [2]. Recent advances in deep learning have led to significant improvements in automated fruit classification. Nonetheless, the empirical implementation of available strategies is constrained by several limitations. Numerous studies have examined small or unbalanced datasets, leading to overfitting and low predictive performance on unseen samples [3], [4]. Moreover, these models tend to be black boxes, which are not easily interpretable and do not provide information about how they make decisions. The high computational cost and incomplete evaluation process, including the lack of cross-validation, learning-curve analysis, and inference-time reporting, further limit the applicability of these methods in real-world settings [5], [6]. To mitigate these shortcomings, this research paper proposes an effective transfer-learning-based system for detecting fruit freshness with high accuracy, interpretability, and efficiency. The methodology employs extensive preprocessing and feature optimization, including image resizing, denoising, contrast enhancement, and normalization, to improve the quality of images

used as inputs to the models. Various state-of-the-art deep learning architectures are evaluated systematically to determine the most effective style. Grad-CAM visualizations are used to improve transparency by highlighting the image regions that contribute most to the model's predictions. Lastly, an extensive evaluation framework, with stratified train-validation-test splits, cross-validation, learning-curve analysis, confusion matrices, and inference-time assessment, will be used to provide a detailed and valid evaluation of model performance.

To overcome the shortcomings of current fruit freshness detection systems, this research contributes to three main aspects:

- 1) This paper presents a transfer learning-based approach to automated fruit freshness detection, combining improved pre-processing and feature enhancement to improve image quality and model performance.
- 2) The methodology incorporates explainable AI approaches to provide transparent predictions and identifies the important regions that drive classification decisions.
- 3) A comprehensive evaluation strategy, including stratified data splitting, cross-validation, and inference-time analysis, ensures reliable and generalizable results suitable for real-world deployment.

The combination of these contributions yields a useful and comprehensible fruit freshness detection system that can be deployed in real agricultural and retail environments.

II. LITERATURE REVIEW

Deep learning has become popular in various applications, including plant disease identification and medical diagnostics, as well as the automatic assessment of fruit quality [7], [8]. Models like CNNs and attention-based networks allow proper disease detection, medical image recognition, and smart food grading systems [9].

This section will be an overview of past literature on the subject of fruit freshness detection with particular attention to new methodologies and main developments. Yue Yuan et al. [5] focus on an innovative approach for detecting the freshness of fruit and vegetables using deep learning. The author proposed a fused CNN-BiLSTM model that extracts features with a CNN and processes dependencies with a BiLSTM. By using the proposed model, they achieved an accuracy of 97.76%. However, the work has notable limitations: cross-validation is not applied, Grad-CAM are not included, and learning curve analysis are missing, limiting robustness and real-world applicability. On the other hand, Harilal et al. [10] introduced the idea of automatic evaluation of fruit freshness with regard to bananas. They trained a GoogLeNet network to extract image features using a pre-trained network. The paper indicates that

the transfer learning model, which is built using GoogLeNet can be effectively used to measure the freshness of bananas, and the model is also helpful in the case of apples. Nevertheless, the work discusses only a limited number of fruit types, lacks cross-validation, Grad-CAM analysis, learning curves, and even the accuracy percentage, making it less transparent about the model's performance, and the reported performance metrics and validation processes are not fully reported. Umer Amin et al. [11] describe a transfer learning algorithm applied to the AlexNet CNN to identify fresh or rotten fruit images. The model architecture includes five convolutional layers and three fully connected layers. Their method was evaluated on three open datasets, achieving accuracies of 98.2%, 99.8%, and 99.3%, respectively. However, the study does not include cross-validation, Grad-CAM analysis, or learning curves, and it does not clearly explain the model's efficiency or the amount of computing power it requires. Additionally, Tapia-Mendez et al. [12] introduce a deep-learning-based system of fruit and vegetable classification and ripeness. The system suggested by the author included two models of transfer learning based on MobileNet V2 architecture, one to recognize 32 classes of fruits and vegetables and one to identify the level of ripeness of 6 classes. The highest accuracy of their model was 97.86% on classification and 100 % on ripeness detection. However, the method is limited in a number of ways. It is based on two distinct models rather than a single architecture, it is not based on Grad-CAM and no cross-validation was utilized. It does not report the parameters count or any deployment viability either. Martinez Pazos et al. [13] discuss two deep-learning-based systems for the classification of the freshness of fruit and vegetable types. The author proposed a transfer-learning model based on the Xception architecture and an Xception-inspired custom CNN model. They applied their transfer-learning model on the proposed model and obtained an accuracy of 98.57%, and the custom CNN of 95.17%. However, with all such promising results, there are few gaps in the study: they do not involve such interpretability tools like Grad-CAM, and they do not include cross-validation as well. Besides this, the paper fails to report the amount of model parameters or the model size in general, and therefore, the considerations related to deploying it in practice remain unexplored. Besides this, the paper fails to report the amount of model parameters or the model size in general, and therefore, the considerations related to deploying it in practice remain unexplored. Similarly, Yuan Shu et al. [14] is about the classification and detection of the fruit freshness by means of deep learning. To promote better fine-grained features extraction, the authors suggest ResNet-101 network with a Non-Local Attention mechanism. The model was highly performing as it reported 94.7% accuracy, which was well above the performance of the conventional CNN models that were being tested. Nonetheless, the model is computationally costly, less accurate, has no cross-validation, and does not explicitly describe the way of training and testing. Yonis Gulzar [15] suggests an approach of classifying fruits known as TL- MobileNetV2 whereby MobileNetV2 is used along with a custom classification head and deep transfer learning. This model was also trained on a large dataset of 26,149 images of 40 types of fruits and reached high accuracy of 99 %, precision of 99 %, recall of 99 %, and F1-score of 99 %. Although the obtained results are impressive, the research is not cross-validated, does not feature explainable AI instruments such as Grad-CAM, or learning curves, preventing the comprehension of the study in terms of its robustness and practical use. Samin Morshed et al. [16] introduces a deep-learning-based method of evaluating fruit quality through a densely

connected convolutional neural network. The author suggested a system that was based on DenseNet201 architecture. Their model was very successful since they could reach up to 99.67 % accurate in all the tasks. Nonetheless, it has several drawbacks, which include: the method has not been cross-validated, the study fails to indicate inference time, the number of parameters involved, whether it can be deployed, and the learning-curve analysis. Prakash Singh [17] trains a system of fruit freshness classification with the help of transfer learning based on the four CNNs: AlexNet, VGG19, GoogLeNet and ResNet-50. AlexNet was the most accurate and also the fastest, and it reached 99.48% validation accuracy. The results are encouraging, but the research is not cross-validated, there is no learning curve, or explainability, like Grad-CAM, which restricts the interpretation of model behavior and real-life performance.

III. METHODOLOGY

This section will explain the methodology that was employed to categorize fruit freshness using deep learning models based on transfer learning. The general workflow will seek to structure and optimize the dataset in an ordered way so that the models will be trained and generate reliable predictions on fresh and rotten fruit images. Figure 1 represents an overview of the whole process.

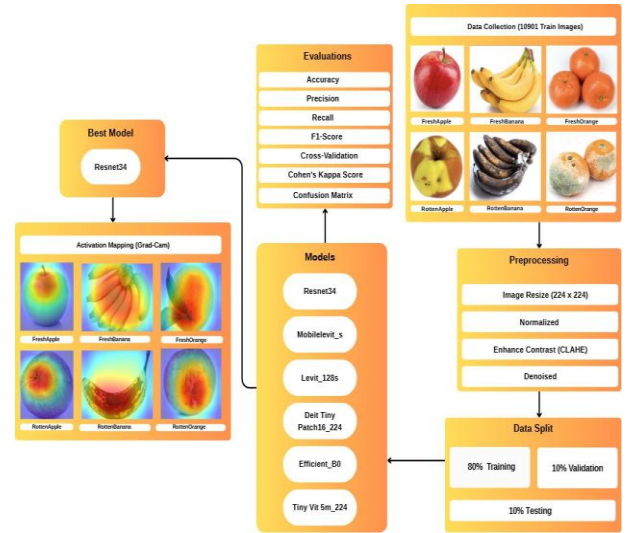


Fig. 1. Workflow of Fruit Freshness Classification

A. Data Collection

In this study, we utilized a dataset collected from Kaggle [18] which consists of 13,599 RGB images that represent both 3 types of fresh and rotten fruits. Images are in all PNG format. The samples of all six classes are represented in Figure 2 to show the visual features of fresh and rotting fruit images.

B. Preprocessing and Feature Enhancement

To ensure efficient model learning, we applied several preprocessing methods to improve the dataset. To begin with, every image was resized to 224×224 pixels to ensure that all the images had a similar input size, which is significant to deep learning models. Then, local contrast was enhanced with the help of CLAHE that helps to distinguish between the slightest details which may state whether some fruit is fresh or starts to rot. Another technique that we also used was to use Non-Local Means filtering technique to get

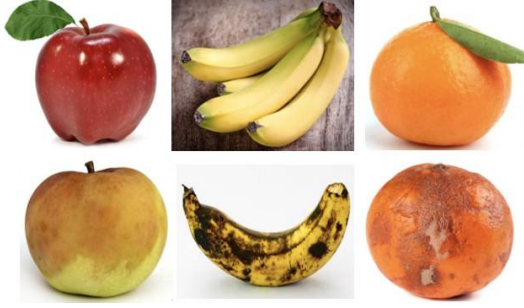


Fig. 2. Sample images from the Fruits Fresh and Rotten dataset: showing fresh and rotten fruits.

our samples of the denoised images and eliminate extra noise without affecting fructose textures that needed to be preserved. Lastly, the images were normalized to the range of 0–1 to make training stable with no extensive fluctuations in the gradient.

C. Quantitative Evaluation of Preprocessing (PSNR and SSIM)

To determine the effectiveness of our preprocessing steps in improving image quality, we compared the images using two standard values, PSNR and SSIM, as reported in Table I. The values of PSNR are significantly high after preprocessing and this indicates that the noise in the images was successfully removed and the general quality of the visuals was greater [19]. Similarly, the increased SSIM scores post-preprocessing prove that the structural information of the pictures was more accurately maintained leading to the image being closer to its clean and high-quality counterparts [20].

TABLE I
PSNR AND SSIM SUMMARY BEFORE AND AFTER PREPROCESSING

Class	PSNR Before	PSNR After	SSIM Before	SSIM After
FreshApples	20.94	26.11	0.3153	0.7184
FreshBanana	21.58	27.09	0.3257	0.8479
FreshOranges	21.59	30.08	0.2104	0.9168
RottenApples	21.55	24.16	0.3051	0.6025
RottenBanana	22.42	24.54	0.2853	0.6734
RottenOranges	21.03	27.20	0.2720	0.6861

D. Dataset Splitting

To prepare the data to be used in model training and evaluation, the dataset was divided into three parts, (80% training, 10% validation and 10% test). A stratified splitting strategy was used to ensure that each of the subsets had equal representation of all the six classes of fruits. The entire images were then separated into training, validation and tests folders which make the dataset well-organized and prepared to execute the model pipeline. A detailed breakdown of the number of images per class in each split is presented in Table II.

E. Baseline Classifiers

In the task of classifying the freshness of fruit, we have chosen six deep learning models to compare their work, to know which architecture is most suitable on our dataset. To begin with, ResNet34 is a widely used convolutional network that incorporates skip connections to improve gradient flow and eliminate vanishing gradient. On the other hand, MobileViT-s introduces a small modular architecture between CNNs and Vision Transformers. It is also efficient and quick inference. Similarly, LeViT-128s is a small model that integrates convolutional networks and attention, achieving fast

TABLE II
DATA SPLIT SUMMARY OF THE FRUIT FRESHNESS DATASET

Class	Train Count	Val Count	Test Count
RottenApples	2355	294	294
RottenBanana	2203	275	276
FreshApples	1670	209	209
RottenOranges	1598	200	200
FreshBanana	1570	196	196
FreshOranges	1483	186	185
Total	10879	1360	1360

inference, and it is very accurate even with the few parameters employed. In addition, DeiT Tiny Patch16-224 is a small Vision Transformer that is trained in a data efficient mode, small patch with minimized self-attention and optimized that can be used to attain high performance without the use of large datasets. Furthermore, EfficientNet-B0 is a CNN architecture that uses compound scaling and achieves accuracy as well as computational efficiency. Lastly, TinyViT-5M-224 is a small Vision transformer with a little bit less than five million parameters, aims to deliver transformer-level performance while remaining lightweight and easy to deploy. Taken together, these baseline models give us a clear point of comparison, helping us understand how well different architectures can identify and classify fruit freshness across all six classes.

F. Model Training Settings

All six base models were trained under a unified configuration to ensure fairness and consistency in comparison. The input image size was set to 224×224 , and the evaluated architectures included ResNet34, MobileViT-s, LeViT-128s, DeiT-Tiny (Patch16-224), EfficientNet-B0, and TinyViT-5M-224. Training was per-

formed using the AdamW optimizer with a learning rate of 3×10^{-4} and a weight decay of 1×10^{-4} , while Cross-Entropy was adopted as the loss function. A batch size of 32 was used for 40 epochs, incorporating early stopping with a patience of 3 epochs. To mitigate overfitting, a dropout rate of 0.5 was applied, and a fixed random seed of 42 was used to ensure reproducibility. All experiments were conducted on Kaggle using dual T4 GPUs with 32 GB RAM and four data loader workers. These standardized settings were selected to maintain a balance between performance and computational efficiency while ensuring consistent training behavior across all architectures.

G. Explainable AI-GRAD CAM

To enhance interpretability, we used Grad-CAM on the most performing model, ResNet34, to visualize the regions of images that guided its freshness prediction. Based on the final checkpoint, the last convolutional layer was used to extract gradients and feature activations to produce class-specific heatmaps.

IV. RESULTS AND DISCUSSION

Classifying fruit freshness across six classes, the ResNet34 model demonstrated strong capability. Five other baselines were compared to model performance in terms of accuracy, precision, recall, F1-score, Cohen’s kappa, AUC and test accuracy with 95% confidence intervals, this guaranteed reliable assessment of the model to be used in practice in the real world.

A. Testing Performances

The performance of six baseline models for fruit freshness classification is summarized in Table III. All models showed high potential in the ability to differentiate between fresh and rotten fruits with high values of accuracy, precision, recall, F1-score, and AUC. One of the models performed relatively low with an accuracy of 0.8978 and F1-score at 0.9009 which means that it has a lower discriminative ability among the six classes of fruits. Of all the models assessed ResNet34 was the highest performer with a test accuracy of 0.9993, precision of 0.9991, recall of 0.9992, F1-score of 0.9991, and AUC of 1.0000. These findings indicate that it is better at extracting discriminative features and is generalizable to all classes. Other models with high performance such as MobileViT-s and EfficientNet-B0 also displayed exquisite classification abilities, although with marginally reduced measures, which prove their strength and dependability.

Altogether, the results suggest that all chosen models were very effective to classify fruit freshness, but ResNet34 exhibited the most reliable, consistent and generalizable findings, so it is the best option to be used in automated fruit quality assessment systems in practice.

TABLE III
COMPARISON OF TEST ACCURACY AMONG EVALUATED MODELS

Model	Accuracy	Precision	Recall	F1-Score	AUC
MobileViT-s	0.9978	0.9977	0.9975	0.9976	0.9998
LeViT-128s	0.9816	0.9828	0.9804	0.9814	0.9992
DeiT Tiny Patch16-224	0.9897	0.9884	0.9897	0.9889	0.9981
EfficientNet-B0	0.9949	0.9940	0.9938	0.9939	0.9998
TinyViT-5M	0.9941	0.9940	0.9945	0.9942	0.9999
ResNet34	0.9993	0.9991	0.9992	0.9991	1.0000

B. Learning Curve Analysis and Cross-Validation of Proposed ResNet34

The learning curves in Figure 3 of the ResNet34 model are demonstrated during the six training epochs, and the improvement in accuracy and loss is obvious and consistent. The training accuracy grows progressively over the first three epochs up to reach 95 %, and the validation accuracy also corresponds to high scores, which are approximately 99-100 %, which demonstrates strong learning and excellent generalization. The loss in training falls quickly in the initial epoch, which is approximately 0.17; the loss in the third epoch is almost 0.04, which shows a rapid acquisition of discriminative features. This also applies to the validation loss, which drops to minimal levels towards the same epoch (around 0.01). This also applies to the validation loss, which drops to minimal levels towards the same epoch (around 0.01). There are minor variations in later epochs, especially an insignificant increase in training loss between epoch four and five. Altogether, the learning curves prove the efficiency of ResNet34 during the training phase and its stable behavior during the training, which justifies its high performance in the classification of freshness of fruits.

The usefulness of the ResNet34 can also be seen in the 5-fold cross validation table below in Table IV. The accuracy of the training is relatively low in all folds, and all folds have a high value of accuracy of over 99 %. The average accuracy of 99.61 also shows that the model is highly generalizable and still achieves high accuracy even when it is trained on other splits of the data, eliminating the chance of overfitting itself to a particular split.

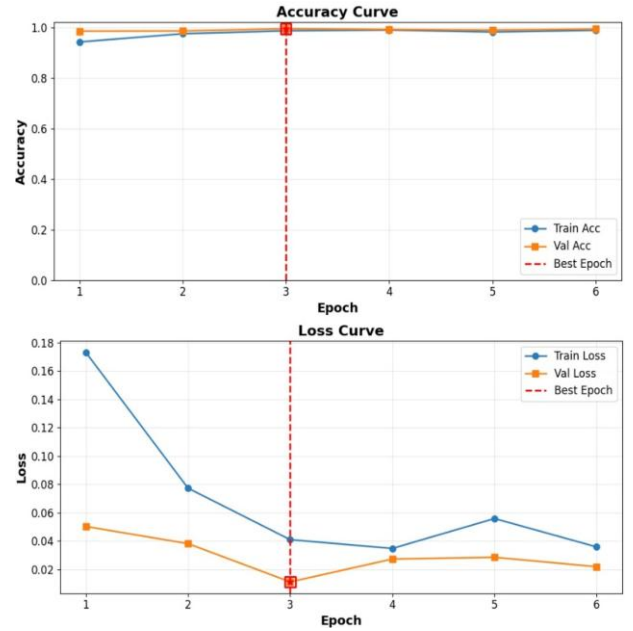


Fig. 3. Learning curves showing the training and validation accuracy and loss of the best-performing model, ResNet34.

TABLE IV
5-FOLD CROSS VALIDATION ACCURACY OF RESNET34

Fold	Training Accuracy
Fold 1	0.9904
Fold 2	0.9996
Fold 3	0.9915
Fold 4	0.9996
Fold 5	0.9993
Mean	0.9961

C. Analysis of Inference Time and Training Time

Figure 4 demonstrates the usage of the GPU and RAM, inference time, and training time of each of the models that have been considered. Despite the lower computational requirements of light models such as LeViT 128s, ResNet34 has superior stability and inference time. Although ResNet34 is a little more expensive in terms of GPU and it also takes a little longer to train, these are not significant in practice. ResNet34 is the most optimal in terms of performance and efficiency in the real world in terms of accuracy, reduced learning behavior, and reduced inference time.

Model	Inference (ms)	Training (s) ↓	GPU (GB) ↓	RAM (GB) ↓
Proposed ResNet34	5.45	391.20	1.56	2.56
DeiT Tiny	10.48	307.92	0.72	2.71
LeViT-	11.34	370.84	0.68	3.36
TinyViT	12.00	1007.02	3.26	3.45
EfficientNet-B0	36.84	277.37	3.07	3.42
MobileViT-S	172.15	481.22	2.97	3.07

Fig. 4. Comparison of inference time, training time, GPU memory usage, and RAM consumption

D. Misclassification Analysis of Proposed ResNet34

Figure 5 shows the confusion matrix of the proposed ResNet34 model, illustrating the effectiveness of the model in classifying the six categories of freshness of fruits. The model is almost perfect in predicting all classes. In the case of fresh apples, fresh bananas, fresh oranges, rotten apples and rotten bananas, all test samples were appropriately identified with no misclassifications. There was only one misclassification in the rotten oranges category, where one sample was wrongly classified as fresh oranges. The rest of the samples of rotten oranges were also correctly classified. In general, the confusion matrix shows that the proposed ResNet34 model has a high level of class separation and minimal confusion between fruit types that are visually similar. Such good performance justifies its accuracy in real world fruit freshness classification problems.

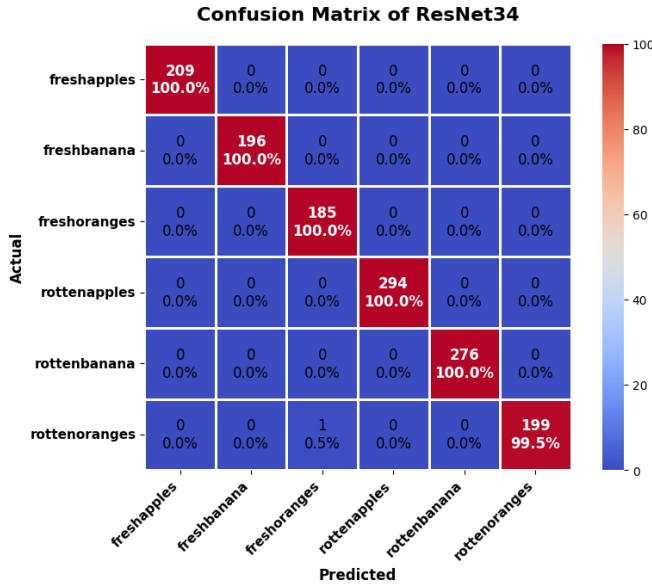


Fig. 5. Confusion matrix of the proposed ResNet34, showing high classification accuracy across all classes with minimal misclassification.

E. Cohen's Kappa and Confidence Interval Analysis

Table V summarizes the Cohens Kappa and the respective confidence interval of all the considered models. The Proposed ResNet34 shows the best Kappa of 0.9991 with a close confidence interval [0.9978,1.0000], which implies a high level of consistency and reliability. EfficientNet-B0 and TinyViT-5M also show high consistency, with Kappa values of 0.9938 and 0.9929. MobileViT-s and DeiT Tiny Patch16-224 have good performances, with each having close confidence intervals. Though LeViT-128s has the lowest Kappa of 0.9778, it also indicates a high level of classification agreement between all the fruit categories.

TABLE V
COHEN'S KAPPA AND CONFIDENCE INTERVAL (CI) FOR ALL MODELS

Model	Kappa	Confidence Interval (CI)
ResNet34	0.9991	[0.9978, 1.0000]
MobileViT-s	0.9973	[0.9953, 1.0000]
LeViT-128s	0.9778	[0.9745, 0.9888]
DeiT Tiny Patch16-224	0.9876	[0.9843, 0.9951]
EfficientNet-B0	0.9938	[0.9910, 0.9987]
TinyViT-5M-224	0.9929	[0.9901, 0.9982]

F. Proposed Model Interpretability using GRAD-Cam Visualization

The Grad-CAM visualizations of the Proposed ResNet34 model are provided in Figure 6. These heatmaps indicate which image parts of both fruit images had the strongest impact on the model predictions. As demonstrated, the model always concentrates on the purely significant areas, including changes in texture, color, and decayed regions that will distinguish fresh samples and rotten samples. This targeted attention confirms that ResNet34 is not only accurate but also interpretable, as it bases its decisions on relevant and clinically important visual features. Altogether, the Grad-CAM outcome supports model reliability as the model highlights the crucial areas that determine fruit freshness.

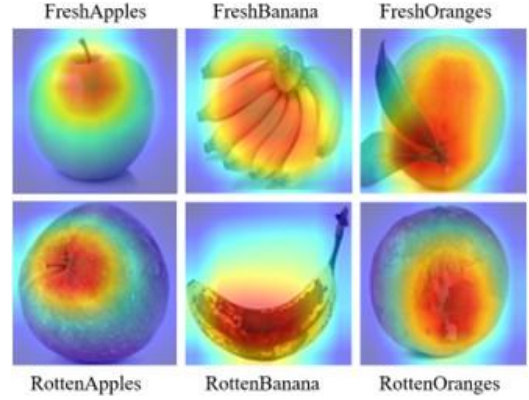


Fig. 6. Grad-CAM visualizations of the proposed ResNet34 model, highlighting class-specific regions in Fruit Classification.

G. Comparison with Previous Studies

Table VI gives a comparative overview of the fruit-freshness classification models that were earlier published and describes the performance and methodological constraints of each model. Yue Yuan et al. [5] achieved an accuracy of 97.76% with a fused CNN-BiLSTM architecture, but the lack of cross-validation, Grad-CAM analysis, and learning-curve analysis restricts the strength of their results. Martinez Pazos et al. [13] achieved 98.57% accuracy with models based on Xception but without Grad-CAM, cross-validation, or model-complexity reporting. Yuan Shu et al. [14] came up with a ResNet-101 with a Non-Local Attention mechanism which achieved an accuracy of 94.7% and had a high computational cost and was not cross-validated. Samin Morshed et al. [16] obtained a high accuracy of 99.67, but they lacked cross-validation, inference-time reporting and learning-curve analysis. In comparison, our proposed ResNet34 achieves 99.93% accuracy while providing Grad-CAM interpretability, full cross-validation, learning curve analysis, confusion matrix and inference-time reporting which countermeasures the major limitations found in the previous studies.

V. CONCLUSION AND FUTURE WORK

This work introduces a detailed transfer-learning model of automated classification of fruit-freshness in 6 categories of fruits: apples, bananas, and oranges. The proposed system compares six current deep-learning architectures, which are ResNet34, MobileViT-S, LeViT-128S, DeiT-Tiny, EfficientNet-B0, and TinyViT-5M, and finds the solution of ResNet34 as the best and the most dependable. Its high-density feature extraction properties, coupled with its powerful preprocessing and stratified data division allow very high quality

TABLE VI
COMPARISON OF PREVIOUS STUDIES AND PROPOSED RESNET34

Study	Model	Accuracy	Limitations
[5]	CNN + BiLSTM Fusion	97.76%	No cross-validation; no Grad-CAM; no learning-curve analysis
[13]	Xception Model	98.57%	No Grad-CAM; no cross-validation; no model-size reporting
[14]	ResNet-101 + Non-Local Attention	94.70%	High computational cost; no cross-validation; incomplete training details
[16]	DenseNet201	99.67%	No cross-validation; no inference-time reporting; no learning-curve analysis
Our Study	Proposed ResNet34	99.93%	None reported; strengths include cross-validation, Grad-CAM, confusion-matrix metrics, and learning-curve analysis

and consistent classification results. The model has a test accuracy of 99.93, a high cross-validation stability, and the largest Cohen k in comparison with all the tested models. Grad-CAM representations additionally validate that the model makes attention to meaningful freshness-related regions therefore, increasing interpretability and transparency. Also, the training and inference analysis demonstrate the ResNet34 has reasonable computational requirements, and this feature makes it appropriate to be applied in the real-world in an agricultural and retail setting.

Although it is well performing, the study has limitations as it uses a dataset that focuses on three types of fruits and this could limit its generalizability in the broader categories of produce. Further research will be aimed at the increase of the dataset to cover additional kinds of fruits, various conditions of the environment, and real-time market situations. Also, the creation of a convenient software platform can encourage feasible implementation in warehouses, supermarkets, and automated sorting lines.

REFERENCES

- [1] S. R. Jaeger, L. Antu'nez, and G. Ares, "An exploration of what freshness in fruit means to consumers," *Food Research International*, vol. 165, p. 112491, March 2023.
- [2] M. E. Haque, F. A. Farid, M. K. Siam, M. N. Absur, J. Uddin, and H. Abdul Karim, "Leafsightx: an explainable attention-enhanced cnn fusion model for apple leaf disease identification," *Frontiers in Artificial Intelligence*, vol. Volume 8 - 2025, 2026. [Online]. Available: <https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2025.1689865>
- [3] Z. F. R. Ahmed, A. K. Abdalla, N. Kaur, and F. Wu, "Insights into recent developments and obstacles in automated fruit ripeness classification," *Green Technologies and Sustainability*, vol. 4, no. 2, p. 100302, April 2026.
- [4] M. N. Absur, K. F. A. Nasif, S. Saha, and S. N. Nova, "Revolutionizing image recognition: Next-generation cnn architectures for handwritten digits and objects," in *2024 IEEE Symposium on Wireless Technology Applications (ISWTA)*, 2024, pp. 173–178.
- [5] Y. Yuan, J. Chen, K. Polat, and A. Alhudhaif, "An innovative approach to detecting the freshness of fruits and vegetables through the integration of convolutional neural networks and bidirectional long short-term memory network," *Current Research in Food Science*, vol. 8, p. 100723, 03 2024.

- [6] Y. Zhang, X. Yang, Y. Cheng, X. Wu, X. Sun, R. Hou, and H. Wang, "Fruit freshness detection based on multi-task convolutional neural network," *Current Research in Food Science*, vol. 8, p. 100733, 2024.
- [7] M. E. Haque, M. H. Saykat, M. Al-Imran *et al.*, "An attention-enhanced cnn ensemble for interpretable and accurate cotton leaf disease classification," *Scientific Reports*, vol. 16, p. 4476, 2026.
- [8] Y. Yuan and X. Chen, "Vegetable and fruit freshness detection based on deep features and principal component analysis," *Current Research in Food Science*, vol. 8, p. 100656, 2024.
- [9] M. N. Absur, "Anomaly detection in biomedical data and image using various shallow and deep learning algorithms," in *Data Intelligence and Cognitive Informatics: Proceedings of ICDICI 2021*. Springer, 2022, pp. 45–58.
- [10] J. Harilal, E. Duggeboina, and P. S. Surekha, "Assessing fruit freshness with deep learning: A case study on banana using googlenet and transfer learning," *Turkish Journal of Computer and Mathematics Education*, vol. 14, no. 3, pp. 1002–1012, 2023. [Online]. Available: <https://www.proquest.com/openview/e70de2de5a6fcb2ca4bd7adae5150073/1?pq-origsite=gscholar&cbl=2045096>
- [11] U. Amin, M. I. Shahzad, A. Shahzad, M. Shahzad, U. Khan, and Z. Mahmood, "Automatic fruits freshness classification using cnn and transfer learning," *Applied Sciences*, vol. 13, no. 14, 2023. [Online]. Available: <https://www.mdpi.com/2076-3417/13/14/8087>
- [12] E. Tapia-Mendez, I. A. Cruz-Albarran, S. Tovar-Arriaga, and L. A. Morales-Hernandez, "Deep learning-based method for classification and ripeness assessment of fruits and vegetables," *Applied Sciences*, vol. 13, no. 22, 2023. [Online]. Available: <https://www.mdpi.com/2076-3417/13/22/12504>
- [13] J. F. Mart'inez Pazos, J. Gul'in Gonza'lez, D. Batard Lorenzo, and A. Orellana Garc'ia, "Freshnets: Highly accurate and efficient food freshness assessment based on deep convolutional neural networks," *Inteligencia Artificial*, vol. 27, no. 74, pp. 48–61, May 2024.
- [14] Y. Shu, J. Zhang, Y. Wang, and Y. Wei, "Fruit freshness classification and detection based on the resnet-101 network and non-local attention mechanism," *Foods*, vol. 14, p. 1987, 06 2025.
- [15] Y. Gulzar, "Fruit image classification model based on mobilenetv2 with deep transfer learning technique," *Sustainability*, vol. 15, no. 3, 2023. [Online]. Available: <https://www.mdpi.com/2071-1050/15/3/1906>
- [16] M. S. Morshed, S. Ahmed, T. Ahmed, M. U. Islam, and A. B. M. A. Rahman, "Fruit quality assessment with densely connected convolutional neural network," *Proceedings of the 2022 12th International Conference on Electrical and Computer Engineering (ICECE)*, pp. 1–4, 2022.
- [17] V. P. Singh, S. N. Talbar, and D. M. Bognulwar, "Sorting of fresh fruits using transfer learning," *Seybold Report*, 2022. [Online]. Available: <https://doi.org/10.5281/zenodo.7259996>
- [18] S. R. Kalluri, "Fruits fresh and rotten for classification," <https://www.kaggle.com/datasets/sriramr/fruits-fresh-and-rotten-for-classification>, 2018, accessed: 2025-12-13.
- [19] Q. Huynh-Thu and M. Ghanbari, "Scope of validity of psnr in image/video quality assessment," *Electronics Letters*, vol. 44, no. 13, pp. 800–801, 2008. [Online]. Available: <https://www.scrip.org/reference/referencespapers?referenceid=2149300>
- [20] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004. [Online]. Available: <https://www.scrip.org/reference/referencespapers?referenceid=1965239>